INSTRUCTIONS TO THE CANDIDATES

- Do not use sketch pens/ high –lighter pens/ color pencils etc., in underlining / highlighting certain points in the answer or while drawing sketches etc.,
- Check the Answer booklet thoroughly. The defective answer booklet may be returned to the invigilator before starting the examination.
- Ensure that the OMR Barcode sheet is properly secured to the answer booklet.
- Do not write on or tamper the Barcode. If you do so, your booklet will not be evaluated.
- The candidate is prohibited from writing his/ her NAME / ADDRESS/ PHONE NUMBER / EMAIL ID / Religion or any other matter on any part of the answer booklet / drawing sheet/ graph sheet/ addressing the examiner in any manner whatsoever in the answer booklet. The candidate is not supposed to tear a page in full or part of the booklet.

  Non-adherence with this instruction shall result in candidate being booked under malpractice.
- Answer should be written on both sides of the paper. Candidate should write in all 25 lines in each page. It is not necessary to begin each answer in a fresh page.
- Candidate should write only question number in the margin.
- Candidate should return the answer booklet to the invigilator before leaving the examination hall.
- **ADDITIONAL SUPPLEMENT OF ANSWER SHEETS WILL NOT BE ISSUED.**

INSTRUCTIONS TO THE EXAMINERS

- Invigilators should ensure that the above instructions are strictly followed by the candidates.
- After verifying the particulars on the OMR sheet, invigilator should sign in the box specified.
- Do not crumple or cut the sides of the sheets.
- Do not write, mark or damage the timing tracks or Barcode.

1

A

Multithreading, it is about the execution of multiple threads in a program. Parallel programming is for the breaking down the program's execution into multiple ~~exo~~ smaller executions for helping the program. Multithreading helps the program to run faster as it shares and ~~divid~~ divides the processor

B

Thread creation.

there are two ways to create a thread :-

1) extending Thread class

2) implementing Runnable interface.

```
class myThread1 extends Thread {
        ~~Pu~~ public void run() {
                System.out.println("Extending Thread class");
        }
}
```

```
class MyThread2 implements Runnable {
        public void run(){
            System.out.print("implementing runnable");
        }
}
```

C

Servlets.

Servlets are java code which resides and run on the server. Generally the client sends a request and the server sends it's response

The server has web.xml (Dependency Discriminator)

With web.XML help Servelts can handle request

D

JDBC.

There are 3 types of statements.

- Statement
- Prepared statement
- Callable statement.

→ Prepared statement:

This statement is used so that a partially complete query can be embedded with it. The rest values can be set by using the setInt(), setString(), e...etc functions

```
Statement stmt = Con.preparedStatement("insert into emp values(?,?)");
                                                       Values(?,?)
stmt.setInt(1, 10);
stmt.setString(2, 'John Doe');
stmt.executeUpdate();
```

→ Callable statement.

This statement is used to call a function which has been userdefined within the database.

2

A

To ensure that only one thread can access the shared resource at a time is by Synchronization.

```
class MyThread extends Thread {
    synchronize public void run() {
        for (int i=0; i<10; i++)
            System.out.println(i);
    }
}
class InSem {
    public static void Main (String[] args) {
        MyThread TA = new MyThread();
        MyThread TB = new MyThread();
        TA.start();
        TB.start();
    }
}
```


B

Producer /Consumer.

The producer and consumer problem is so that the consumer gets and the producer puts. Without each ones request the other's response is vain.

So, when the two threads producer and consumer are in execution. they both need to be in sync. While one does it's job the other stops.

This can be acieved by the wait() and notify() methods. While one thread goes into wait(). the other does it's process, after it's process it notifies the thread with notify() method and it goes into wait(), vice versa. and so on till the whole program execution is completed.

∴ This is how the wait() and notify() methods can be used to coordinate the interactions between Producer and Consumer.

C

JDBC

There are 5 canical steps in JDBC

-> Register the driver.

-> Get the connection

-> Create statement object

-> Execute queries

-> Close the connection.

-> Register the driver.

Acknowledging which database model we are going to connect. MongoDB, PostgresqL, MySQL

//Class. ForName ("<Drivers name>");

Class. ForName ("Org. Postgresql. Driver");

-> Get the Connection

Estabiling connection with the database

Connection con = DriverManager.get Connection (
"<url>", "<username>", "<pass>");

("url", "Username", "password")

-> Create Statement object.

Creation of object for the queries to be exeated on.

~~Connection~~

Statement stmt = con. Statement();

-> Execute queries

Execute your DBMS queries through our Statement object.

~~Stm~~ stmt. execute("<DBMS queries>");

-> close Connection

Close the DBMS connection

con. close ();

D

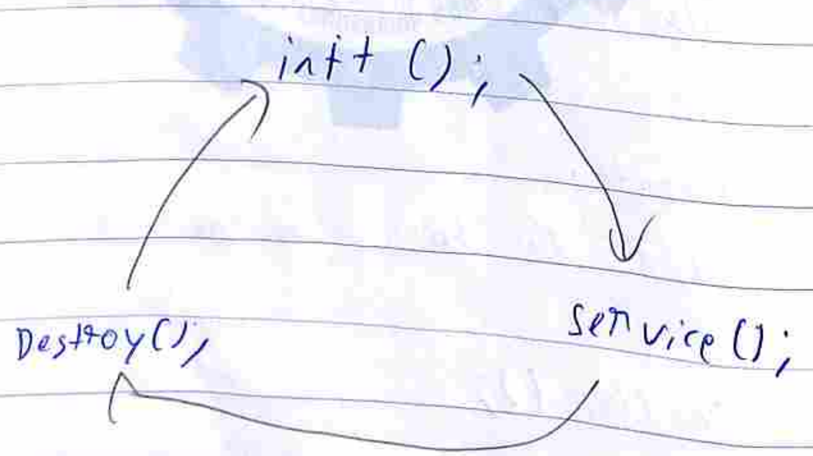
Servlet.

init → initalization of servlet.

service → Program Process Stored for Execution

Destroy → Destroy (close) servlet.

Life cycle



3

AB

```
class MyThread extends Thread {
    String name;
  public MyThread (String t) {
        this. name = t;
    }
    public void run() {
        System.out.println(name);
    }
}
class InSem {
    public static void main (String[] args) {
        MyThread T1 = new MyThread ("Rose");
        MyThread T2 = new MyThread ("Jisoo");
        MyThread T3 = new MyThread ("Jimin");
        MyThread T4 = new MyThread ("JungKook");
        MyThread T5 = new MyThread (" Jennie ");
        MyThread T6 = new MyThread ("BlackPink");
```

```
T1. set Priority ( 8 );
T2. set Priority ( 5 );
T3. set Priority ( 4 );
T4. set Priority ( 6 );
T5. set Priority ( 2 );
T6. set Priority ( 10 );

T1. Start ();
T2. Start ();
T3. Start ();
T4. Start ();
T5. Start ();
T6. Start ();
}
}
```

O/P

= Black Pink
= Rose
= Jungkook
Jisso
Jimin
Jennie

A.
Producer / consumer

This problem is so that consumer gets and the producer puts.

When the two threads producer and consumer are in execution they both need to coordinate and ~~function~~.. function.

For that coordination we use the wait() and notify() methods.

While one waits, other procedes. After its done it notifies the other one and it falls into wait. This coordination helps the producer and consumer.

4

A

```
import Java.io.*
import Java.Sql.*;

public class Insem {
    public static void main(String[] args){

        class.ForName("org.Postgresql.Driver");

        String url = " ";
        String usr = " ";     varies with individual DB
        String Psd = " ";

        Connection Con = DriverManager.getConnection(url, usr, Psd);
        statement Stmt = Con.statement();

Create -> Stmt.execute(" Create table emp (int id, varchar(30) name)");

Insert -> Stmt.execute(" Insert into emp values (1, 'Satin');");

Read -> Result Set RS= Stmt.execute ("select * from emp;");
        System.out.println(RS);

Delete -> Stmt.execute(" Drop table emp; ");

        Stmt.close();
        Con.close();
    }
}
```


B

Fibonacci Sequence

$n = (n-1) + (n-2)$

The number is equal to the sum of the previous two numbers.

$n = 5$

$\rightarrow$ 1 1 2 3 5

NO TIME